

ETHIOPIAN HEALTH PROFESSIONALS LICENSING EXAMINATION(EHPLE) PRACTICE TEST BOOKLET 2025

PHARMACY



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MINISTRY OF HEALTH-ETHIOPIA

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HEALTHIER CITIZENS FOR PROSPEROUS NATION!



Message from the State Minister, Ministry of Health – Ethiopia



Mrs. Frehiwot Abebe

Improving the quality of healthcare remains a critical global priority and is essential for achieving sustainable development and universal health coverage. One of the most effective ways to ensure high standards in healthcare delivery is through the consistent assessment of professional competencies. In line with this commitment, the Ministry of Health of Ethiopia launched the Ethiopian Health Professionals Licensing Examination (EHPLE) in 2019 for undergraduates in seven core health disciplines. Since its inception, the program has expanded to encompass 17 health professions, reflecting our continued effort to build a competent and responsive health workforce.

The EHPLE was established with the primary aim of identifying and certifying health professionals who possess the minimum competencies required to provide safe and effective care. This ensures that all practicing professionals are equipped to meet the expectations of the communities they serve. The exam is coordinated by the Health and Health-Related Institutions and Professionals' Regulatory Lead Executive Office (LEO), which consists of four specialized desks. The LEO plays a vital role in system strengthening, enabling the exam to be conducted with increased efficiency, organization, and quality assurance.

It is important to recognize that the EHPLE is not a conventional academic or recruitment examination. Rather, it is a rigorous, competency-based assessment designed to evaluate the practical skills and critical thinking abilities that are essential for professional practice. This *Practice Items Booklet* is part of our broader strategy to support examinees and educators in preparing for the licensing process. It offers a realistic overview of the exam structure, promotes a deeper understanding of its objectives, and fosters transparency and sustainability in implementation.

The development of this booklet has been made possible through the collaborative efforts of experts from higher education institutions, the Addis Ababa University Testing Center, and professional associations. Their insights and dedication have been instrumental in shaping this resource. I extend my heartfelt thanks to all contributors and to the Ministry of Health team for their tireless work and steadfast commitment to excellence in healthcare regulation.

We believe this booklet will serve as a valuable tool for candidates preparing for the examination and for educators guiding them toward success. Together, we are building a stronger, more competent health workforce for Ethiopia.

Introduction EHPLE

Welcome to the 2025 Edition of the Ethiopian Health Professionals Licensing Examination Practice Test Booklet.

This resource is designed to support candidates in their preparation for the national licensing examination, ensuring they are well-equipped with the knowledge and clinical reasoning skills required to deliver competent and ethical healthcare across Ethiopia.

In Ethiopia, the licensing examination serves as a pivotal tool in evaluating the competency of health professionals who have completed their academic training and are ready to transition into independent practice. The exam not only ensures public safety by verifying professional readiness but also upholds national and international standards in healthcare service delivery. As the healthcare system continues to expand and evolve, a strong, well-regulated professional workforce becomes even more essential.

This practice booklet contains 200 multiple-choice questions modelled after the structure, content, and format of the actual licensing examination. The questions span across core clinical domains essential for entry-level practice. This includes areas such as Patient care, leadership and management, ethics, and Health Promotion as outlined in the Information Booklet 2024.

The booklet is intended to:

- Familiarize candidates with the structure and cognitive level of the actual licensing examination.
- Offer opportunities for self-assessment and identification of strengths and gaps in knowledge.
- Encourage critical thinking and clinical decision-making in scenarios reflective of real-world healthcare practice in Ethiopia.
- Reduce exam-related anxiety by providing realistic practice experiences.

Administered under the auspices of the Ministry of Health and relevant regulatory bodies, the Ethiopian Health Professionals Licensing Examination is grounded in evidence-based frameworks and aligned with both national needs and global best practices. It has been structured to assess a range of competencies including medical knowledge, communication, professionalism, and safe clinical practice.

This practice test is not a predictive tool, and performance on it does not guarantee similar outcomes on the actual exam. However, it provides an invaluable opportunity for familiarization with question styles, pacing strategies, and content review. We strongly recommend that candidates simulate real testing conditions while using this resource, including timing their responses and reviewing rationale for both correct and incorrect answers.

The goal of licensing examinations is not only to regulate entry into the profession but also to inspire excellence and lifelong learning. By engaging with this practice test, you are taking a proactive step toward professional development, accountability, and a commitment to serving your patients with competence and compassion.

We wish you success on your licensing journey—and beyond.

Acknowledgements

This *Practice Test Items Booklet for the Ethiopian Health Professionals Licensing Examination* is the result of a collaborative effort among dedicated professionals committed to advancing the quality of health sciences education in Ethiopia. The Ministry of Health extends its sincere appreciation to all those who contributed their time, expertise, and insights to the development of this essential resource.

Special recognition is given to the item developers and reviewers from various higher education institutions—both public and private—whose academic rigor and practical experience were instrumental in shaping the content and ensuring its relevance to real-world healthcare practice. Their contributions have helped create a high-quality, standardized practice tool that reflects national competencies and global benchmarks.

We are especially grateful to the team at the Addis Ababa University Testing Center, whose technical support and academic guidance were critical throughout the process. Their leadership in assessment design and educational evaluation has significantly strengthened the quality of this booklet.

We also thank the committed staff of the Ministry of Health, particularly those within the Health and Health-Related Institutions and Professionals' Regulatory Lead Executive Office (HHRIPR LEOs), whose coordination, oversight, and unwavering dedication ensured the successful completion of this project.

Together, these collective efforts reflect our shared vision of building a competent, ethical, and resilient health workforce for Ethiopia.

Pharmacy

1. A 30-year-old known HIV patient who is on ART came to a clinic with a complaint of severe headache, fever, and weight loss of two weeks duration. On physical examination, there is a sign of lymphadenopathy and nuchal rigidity.

What is the most likely important history that should be taken during the pharmaceutical care process?

- A. Allergy
- B. Adherence
- C. Alcohol intake
- D. Sexual behavior

ANSWER: B

2. A 28-day-old male patient was admitted to a neonatal intensive care unit with a diagnosis of bacterial meningitis. His current medications include; ceftriaxone and vancomycin calculated based on his age and weight. Culture was not done and treatment was started empirically.

What is the most likely drug therapy problem?

- A. Needs additional drug therapy
- B. Inappropriate drug selection
- C. Unnecessary drug therapy
- D. Adverse drug reaction

ANSWER: A

3. An eight-year-old male patient came to a dermatology clinic accompanied by his mother. On examination, she has a central clearing surrounded by an advancing red, scaly, elevated border. A KOH test was done and the diagnosis is Tinea capitis. His current medication is griseofulvin 250mg daily for 6 weeks.

What is the most appropriate non-pharmacologic intervention for this patient?

- A. Avoid fatty food
- B. Take the medicine with fatty food.
- C. Take the medicine before food
- D. Take the drug on empty stomach.

ANSWER: B

4. A 62-year-old male patient was admitted to a surgery ward to undergo gastroduodenal surgery. One hour before the surgery, he was given ceftriaxone 1gm IV and the treatment continued for five days after the surgery.

What is the most appropriate measure to be taken for this condition?

- A. Recommending to add neomycin
- B. Recommending to stop ceftriaxone
- C. Replacing ceftriaxone with cefazolin
- D. Recommend to increase the dose of ceftriaxone to 1g BID

ANSWER: B

5. A 23-year-old newly diagnosed HIV patient came to the ART clinic to take her medication after getting adequate counseling from the health care provider. Her laboratory result shows a CD4 count of 450 cells/mm³.

What is the most preferred first-line regimen for this case?

- A. AZT (or ABC) + 3TC + RAL
- B. TDF + 3TC (or FTC) + EFV
- C. TDF + 3TC (or FTC) + DTG
- D. ABC + 3TC + DTG

ANSWER: C

6. A 70-year-old male patient presented to an emergency department with a chief complaint of fever and headache for four day duration. On physical examination, he has nuchal rigidity, and positive Kerning's and Brudzinski's signs. Finally, he is diagnosed with acute bacterial meningitis.

What is the most appropriate initial step in the management of this patient?

- A. Administering IV antibiotics after taking a sample
- B. Doing antibiotic susceptibility test
- C. Giving steroids like prednisolone
- D. Identification of pathogen

ANSWER: A

7. A 65-year-old female living with HIV and on ART for the last 10 years is presented with a chief complaint of persistent diarrhea and vomiting of two weeks duration. She was also treated for toxoplasmosis one month ago. Up on investigation, the CD4 count is persistently low and below baseline, viral load was measured twice and it is more than 20,000 copies/ml in both tests. Finally, treatment failure is confirmed.

What is the next most appropriate step in the management of this patient?

- A. One drug that is suspected of resistance can be replaced by a new one
- B. Sending her with the same regimen and advising her to be adherent
- C. A new regimen should be started with at least two new drugs
- D. A regimen that has failed in the past can be reinitiated

ANSWER: C

8. A 40-year-old female patient was admitted to an internal medicine ward with a diagnosis of stage four retroviral infection defined by cryptococcal meningitis. What is the most appropriate initial step in the management of this patient?

- A. Starting antiretroviral drugs first followed by antifungal drugs
- B. Starting both antiretroviral drugs and antifungal drugs at the same time
- C. Initiating antifungal drugs followed by antiretroviral drugs within two weeks
- D. Initiating antifungal drugs followed by antiretroviral drugs after four weeks

ANSWER: D

9. A 22-year-old male student was brought to a psychiatric OPD of a hospital by his friends after they found him talking and laughing alone. He becomes aggressive while communicating with people. On mental status examination, he has decreased and disorganized speech.

What is the most appropriate history for pharmaceutical care in this patient?

- A. Past medication history
- B. Past chronic medical history
- C. Past laboratory investigation results
- D. Current and past psychiatric history

ANSWER: A

10. A 45-year-old female patient newly diagnosed with major depressive disorder came to an OPD pharmacy after she took fluoxetine 20mg for one week. She is not happy with this management because her problem is not resolved and she requested to change the medication.

What is the next most appropriate step in the management of this patient case?

- A. Increasing the dose of fluoxetine
- B. Switching to another antidepressant
- C. Combining with another medication
- D. Counsel to take the medicine for extra two weeks and to come back again

ANSWER: D

11. A 40-year-old male patient is presented to a neurologic clinic with chief complaints of abnormal body movement and loss of consciousness. Vitals signs are normal and EEG shows seizure.

What is the most relevant history for pharmaceutical care in this case?

- A. Frequency of seizure events and precipitating factors
- B. History of epilepsy of his wife
- C. Past medical history of the patient
- D. Family history of epilepsy

ANSWER: A

12. A 30-year-old female patient was diagnosed as having epilepsy with generalized tonic-clonic type of seizures. She was treated with 100mg daily carbamazepine for the last three months. Currently, she is presented with multiple episodes of seizures.

What is the most likely problem that is related to drug therapy in this patient?

- A. Dose too low
- B. Ineffective drug therapy
- C. Unnecessary drug therapy
- D. Needs additional drug therapy

ANSWER: A

13. A 55-year-old female patient was diagnosed with stage four breast cancer. She is admitted to an internal medicine ward to get palliative care. She is on chemotherapy and morphine. Recently morphine dose was escalated to 10mg mg BID to 10 mg TID. After that, she experienced excessive sedation, was not able to be aroused with sternal stimulation, and have a significant decline in breathing.

What is the most likely drug therapy problem in this patient?

- A. Adverse drug reaction
- B. Ineffective drug therapy
- C. Unnecessary drug therapy
- D. Needs additional drug therapy

ANSWER: A

14. An 18-year-old male patient came to an emergency department after having idiopathic epilepsy with generalized tonic-clonic seizures. The two seizures occurred without recovery of consciousness. He fulfilled the criteria and was diagnosed with status epilepticus.

What is the most appropriate initial step in the management of this case?

- A. Start diazepam
- B. Give carbamazepine
- C. Start with phenobarbital
- D. Administer Valproic acid

ANSWER: A

15. A 28-year-old mother gave birth at a hospital and her baby was born with a neural tube defect. She never visited a hospital throughout her pregnancy and she didn't take any medication. She also doesn't have any chronic known medical illness.

What is the most likely cause of the neural defect in this case?

- A. Iron deficiency
- B. Medication toxicity
- C. Labour complication
- D. Folic acid deficiency

ANSWER: D

16. A 28-year-old primigravida is admitted to the labor and delivery suite for labor induction. She is at 33 weeks of gestation and ultrasound has a normal obstetric examination. She is taking misoprostol 25 micrograms for cervical ripening and labor induction. Twelve hours after administration of two 25 microgram doses of misoprostol, her cervix has responded, but she has not developed a consistent pattern of uterine contractions.

What is the most likely drug therapy problem in this case?

- A. Needs additional drug therapy
- B. Unnecessary drug therapy
- C. Ineffective drug therapy
- D. Dose too high

ANSWER: A

17. A 23-year-old female with mild hair growth above her upper lip, mild acne, and a history of irregular menstrual periods came to a gynecology OPD. Since age 12, she has had six to nine periods per year at intervals that vary from 30 to 70 days. She has the desire to get pregnant. Her signs and symptoms are in line with polycystic ovarian syndrome.

What is the most appropriate treatment in this case?

- A. Metformin
- B. Spironolactone
- C. Clomiphene citrate
- D. Combined oral contraceptives

ANSWER: A

18. A 45-year-old known diabetic patient came to his primary care provider with a chief complaint of failure to achieve a penile erection appropriate for satisfactory sexual intercourse. The clinician prescribed sildenafil.

What is the most appropriate monitoring action to ensure the safety of the medication?

- A. Clinical symptoms of erectile dysfunction
- B. Hearing/visual loss
- C. Lipid profile
- D. Electrolytes

ANSWER: B

19. A 35-year-old pregnant woman came to a hospital for her first prenatal visit. She is a known hypertensive patient on follow-up and has a history of gestational diabetes mellitus. A nurse wants to screen her for diabetes mellitus.

What is the most appropriate screening test for this patient?

- A. Hemoglobin A1C
- B. Fasting blood glucose
- C. Random blood glucose
- D. Oral glucose tolerance test

ANSWER: D

20. A 19-year-old female came to a hospital with a chief complaint of having trouble sleeping despite the use of over-the-counter sleep aids. Physically, she has warm, moist skin and appears slightly anxious but responds appropriately to all questions. Vital Signs show a pulse rate of 92 beats/min, blood pressure of 108/73 mmHg, respiratory rate of 12 breaths/min, and temperature of 37 degrees centigrade. Laboratory findings show; TSH of 0.15 mIU/L (reference range 0.4–4.5 mIU/L), FT4 of 2.2 ng/dL (reference range 0.7–1.9 ng/dL), and + TSHR-SAbs. Electrolytes and CBC are normal.

What is the most likely diagnosis of this patient?

- A. Hyperthyroidism
- B. Myxedema coma
- C. Hypothyroidism
- D. Thyroid storm

ANSWER: A

21. A 64-year-old man complains of gastrointestinal symptoms like loose stools secondary to metformin he is taking for type two diabetes. Because of this, he stopped the medication. He was also diagnosed with hypertension and hyperlipidemia. He is obese with a BMI of 32 kg/m². Laboratory results for liver and renal function tests are within normal limits. Current medications include lisinopril and atorvastatin. What is the next most appropriate option for the treatment of this patient?

- A. Insulin
- B. Glyburide
- C. Liraglutide
- D. Pioglitazone

ANSWER: C

22. A 34-year-old woman with a 10-year history of type one diabetes mellitus is planning to become pregnant and needs health professional advice about her pre-conception care and diabetes. During the consultation, the health professional discovered that she has been taking folic acid five-mg daily for the last six months. Her most recent HbA1c is 7.3 percent. Her regular medications are lisinopril 10 mg daily, simvastatin 40 mg daily, and NPH insulin 20/10 IU twice daily.

What is the most appropriate advice for this patient?

- A. Target HbA1c of less than 7.5 percent before conceiving
- B. Discontinue folic acid upon conception until she starts the second trimester
- C. Everything is okay and no change in your current therapy is needed
- D. Discuss with a doctor to change lisinopril and simvastatin

ANSWER: D

23. A 76-year-old woman with a 15-year history of type two diabetes and a five-year history of mild renal dysfunction complains of feelings of lethargy and sleepiness. Her blood glucose is 46 mg/dL. Glycaemic control was good for the past several months with glyburide 10 mg twice daily, metformin 850 mg three times daily and moderate-intensity exercise five days per week. Recently, she has begun fasting. What was the most likely predisposing factor for hypoglycemic episodes in this patient?

A. Fasting
B. Old age
C. Renal dysfunction
D. Moderate intensity exercise

ANSWER: A

24. A 46-year-old woman with hypertension on hydrochlorothiazide 25 mg daily came to OPD for a follow-up appointment. She complains of fatigue and occasional headaches. On physical examination, her blood pressure is 146/84 mmHg. Pertinent laboratory findings are HgA1c 7.9 percent and fasting plasma glucose 178 mg/dL. The patient is diagnosed with type two diabetes mellitus.

What is the most appropriate initial therapy for her diabetes?

A. Comprehensive lifestyle modifications
B. Lifestyle modification plus metformin
C. Lifestyle modification plus sulfonylurea
D. Lifestyle modification plus thiazolidinedione

ANSWER: B

25. A 23-year-old man reports an exacerbating heartburn and regurgitation. He has been self-medicating with omeprazole 20 mg before his breakfast for the last two weeks. But, nighttime symptoms are still not improved. His weight is 80 kg, and his height is 180 cm. He has no known drug allergies.

What is the next most appropriate step in the care of this patient?

A. Protein restriction
B. Avoidance of trigger food items
C. Caloric restriction to lose weight
D. Addition of omeprazole before the evening meal

ANSWER: D

26. A 42-year-old male presented to a hospital with complaints of dyspepsia and Epigastric pain that typically occurs one to three hours after a meal and is often relieved by food. He denies drug use in the past three months. Testing for *Helicobacter pylori* infection is planned as he is suspected with active peptic ulcer.

What is the most appropriate diagnostic test for this patient?

A. Complete Blood count
B. Urea breath test
C. Serologic testing
D. Endoscopic test

ANSWER: B

27. A 58-year-old man with a long-standing history of chronic liver disease is admitted to a hospital with a two-week history of vomiting, increased abdominal distension, and worsening jaundice. His medical history shows a history of ascites secondary to portal hypertension six months ago. He is taking medications; spironolactone 100 mg orally daily and furosemide 40 mg orally daily. Random spot urine electrolyte concentration shows; sodium 10 mEq/L and potassium 28 mEq/L.

What is the most likely explanation for this finding?

- A. Sodium or water restriction is inappropriate
- B. Over diuresis with severe hypo-natremia
- C. The need to discontinue diuretic therapy
- D. The need for diuretic dosage titration

ANSWER: D

28. A 25-year-old pregnant woman came to a hospital with complaints of persistent nausea and vomiting, despite she adheres to lifestyle modifications. She shows signs of dehydration and weight loss consistent with a diagnosis of hyperemesis gravidarum. Currently, she is in her 8th week of gestation. What is the most appropriate option in the management of this patient?

- A. Ondansetron
- B. Promethazine
- C. Metoclopramide
- D. Methylprednisolone

ANSWER: A

29. A 63-year-old man with a known history of chronic liver disease has been treated for ascites. For the last few days, he has been taking spironolactone 400 mg/day and furosemide 80 mg orally twice daily with sodium restriction. But there is no major improvement in diuresis and he developed shortness of breath progressively.

What is the next most appropriate therapeutic measure for this patient?

- A. Addition of angiotensin-converting enzyme inhibitors
- B. Increasing the dose of spironolactone and furosemide
- C. Large-volume paracentesis
- D. Addition of β -blockers

ANSWER: C

30. A 60-year-old woman is prescribed alendronate 5 mg orally daily for the Prevention of postmenopausal osteoporosis. What is the most appropriate intervention to prevent esophageal irritation?

- A. Use delayed-release formulation of risedronate
- B. Instruct her to take her medication with calcium or vitamin D
- C. Instruct her to take her medication with a glass of water after her breakfast
- D. Instruct her to stay fully upright for at least 30 minutes after taking alendronate

ANSWER: D

31. A 28-year-old woman with a BMI of 22.8 kg/m² is worried about developing osteoporosis. Her 75-year-old grandmother had osteoporosis and recently her mother (age 53) was told that she is at increased risk for osteoporosis. Her diet typically consists of one cup of skim milk for her breakfast. What is the most important risk factor for developing osteoporosis in this case?

- A. A positive family history
- B. Low body mass index
- C. A low calcium intake
- D. Female gender

ANSWER: A

32. A 55-year-old man who had noticed red lesions that flaked covering his back and elbows was diagnosed with plaque psoriasis. He started clobetasol 0.1 percent ointment and phototherapy and has been doing well for several months. Today, he reports that his skin looks better, but he feels joint pain.

What would be the most appropriate treatment for this patient at this time?

- A. NSAIDs
- B. TNF-alpha inhibitors
- C. Systemic glucocorticoids
- D. Disease-modifying anti rheumatic drugs

ANSWER: A

33. A five-month-old infant is brought to a pediatric clinic with a rash that is erythematous with poorly defined edges in the folds of the skin around the diaper area. Other typical symptoms include irritation and pruritus. The past medication history is not contributory.

What is the most appropriate first-line therapy for this infant?

- A. Zinc oxide paste
- B. Clotrimazole cream
- C. Hydrocortisone cream
- D. Tetracycline ointment

ANSWER: A

34. A 65-year-old male client was diagnosed with prostate cancer. He has no history of co-morbidities and smoking. He usually exercises regularly and eats a diet consisting mainly of vegetables and fruit four to five times a week. His elder brother and father are alive with no history of chronic disease.

What is the most important risk factor for the prostate cancer in this case?

- A. Age
- B. Diet
- C. Lifestyle
- D. Family history

ANSWER: A

35. A 68-year-old, 59-kg man is diagnosed with early-stage (stage II) non-small-cell lung cancer (NSCLC). Adjuvant chemotherapy of carboplatin targeted to an AUC of 6 mg/mL $\times$ minute and paclitaxel 135 mg/m² is initiated.

What is the most predictable severe toxicity that would be expected to happen with this regimen?

- A. Alopecia
- B. Cardiotoxicity
- C. Nephrotoxicity
- D. Myelosuppression

ANSWER: D

36. A 46-year-old man came to a hospital because he had noticed that his urine was unusually dark a few weeks ago. He is pale but does not appear jaundiced. Rectal examination revealed black stools. Laboratory findings reveal low hemoglobin, a normal alanine transaminase, a slightly raised total bilirubin and reticulocyte percentage of 13.5 percent.

What laboratory findings are consistent with hemolytic anemia in this scenario?

- A. An increased reticulocyte percentage
- B. A normal alanine transaminase
- C. Low hemoglobin
- D. Black stools

ANSWER: A

37. A 65-year-old man was diagnosed with deep vein thrombosis and started unfractionated heparin 17,500 IU SC twice daily. A follow-up CBC five days later reveals a platelet count of $95,000/\text{mm}^3$, baseline $255,000/\text{mm}^3$.

What is the most likely drug therapy problem in this scenario?

- A. Dose too high
- B. Adverse drug reaction
- C. Ineffective drug therapy
- D. Unnecessary drug therapy

ANSWER: B

38. A 60-year-old man came to OPD for a general check-up. He is worried about his health condition due to a family history of cardiovascular disease. He has a history of alcohol consumption and Salbutamol puff as a means for his asthma. On physical examination, his blood pressure was 145/90 mmHg and other results were normal.

What is the most relevant modifiable history for pharmaceutical care in this case?

- A. Age
- B. Alcohol use
- C. Family history
- D. Asthma comorbidity

ANSWER: B

39. A 76-year-old woman with heart failure and type two diabetes mellitus was admitted to medical ward due to a new onset of non-valvular atrial fibrillation and prescribed warfarin.

What is the most likely medical history that justifies the initiation of warfarin for this patient?

- A. Prevention of stroke
- B. Ischemic heart disease
- C. Prevention of ventricular fibrillation
- D. Prevention of chronic kidney disease

ANSWER: A

40. A 30-year-old man came to OPD with chief complaint of left leg pain and swelling of the popliteal region for two days duration. He had history of strain due to fall down, otherwise he is healthy. He has no pertinent family, medication, and medical history. His calf swelling of the left leg was one centimeter higher than the right and laboratory findings showed increased WBC and a negative D-dimer test.

What is the most likely interpretation of laboratory results in this case?

- A. No thrombosis
- B. Arterial thrombosis
- C. Pulmonary embolism
- D. Deep venous thrombosis

ANSWER: A

41. A 65-year-old woman came to a private clinic for an echocardiography study. She is worried about her diabetes complications and her family history of cardiovascular disease, otherwise, she is healthy. She has a medical-history of well-controlled type two diabetes for the last three years. The cardiologist told her that her echocardiography and physical examination results were normal. However, her BMI is 32 kg/m^2 .

What is the most likely diagnosis in this case?

- A. Stage A heart failure
- B. Stage B heart failure
- C. Stage C heart failure
- D. Stage D heart failure

ANSWER: A

42. A 32-year-old woman with chronic heart failure and hypertension came to a clinic for routine follow-up. She has been on metoprolol, spironolactone, and amlodipine. The baseline laboratory data are unavailable. But, today her serum creatinine is 2.5 mg/dl and serum potassium is 6 mEq/L .

What is the most likely drug therapy problem in this case?

- A. Non-compliance
- B. Adverse drug reaction
- C. In-effective drug therapy
- D. Unnecessary drug therapy

ANSWER: B

43. A 78-year-old man was admitted to an intensive care unit after diagnosis of ST-segment elevated myocardial infarction. The man arrived at the hospital after 48 hours of signs and symptoms. Following his admission, the team put him on nitro-glycerine, Alteplase, metoprolol and intranasal oxygen immediately, and ordered additional investigations.

What is the most likely drug therapy without indication in this case?

- A. Alteplase
- B. Metoprolol
- C. Nitro-glycerine
- D. No medication without indication

ANSWER: A

44. A 72-year-old woman with ischemic heart disease, bilateral renal artery stenosis, and heart failure came to a clinic for routine follow-up. She has been on aspirin, atorvastatin, metoprolol, and spironolactone. The baseline laboratory data on the day of these medications' initiation showed; serum creatinine of 2.5 mg/dl and serum potassium of 6 mEq/L .

What is the most likely drug therapy problem identified in this case?

- A. Drug without indication
- B. In-effective drug therapy
- C. Drug with contra-indication
- D. Need additional drug therapy

ANSWER: C

45. An 80-year-old man with hypertension came to an emergency department with a sudden loss of consciousness and left-side hemiparesis of four hours duration. He has no pertinent history other than the discontinuation of his anti-hypertensive medication for the last two weeks. CT scan shows ischemic stroke and his blood pressure is 150/90 mmHg.

What is the most appropriate immediate intervention in this case?

- A. Alteplase
- B. Labetalol
- C. Phenytoin
- D. Nifedipine

ANSWER: A

46. A 56-year-old man was admitted to a medical ward with a new onset of heart failure. On physical examination, his blood pressure is 100/72mmHg, has bilateral grade two edema and dullness on auscultation with orthopnea of three pillows. He has a normal renal function.

What is the most appropriate management for this patient?

- A. Initiate furosemide, metoprolol, and enalapril immediately
- B. Initiate furosemide immediately and metoprolol and enalapril at discharge
- C. Initiate furosemide and metoprolol immediately and enalapril at discharge
- D. Initiate furosemide and enalapril immediately and metoprolol at discharge

ANSWER: D

47. A 45-year-old man was admitted to an emergency department due to an ischemic stroke. His blood pressure is 200/100 mmHg. He arrived at the emergency after three hours of symptom onset. He has no known contra-indication to fibrinolytic therapy, thus, Alteplase was planned for this patient.

What is the immediate blood pressure goal before initiating Alteplase in this case?

- A. less than 220/120
- B. less than 185/110
- C. less than 140/90
- D. less than 130/80

ANSWER: B

48. A 42-year-old woman with chronic heart failure and atrial fibrillation came to a clinic for routine follow-up. She has been on metoprolol, digoxin, and furosemide with good adherence. Vital signs showed a blood pressure of 120/70 mmHg and pulse rate of 80 beats per minute.

What is the most appropriate monitoring parameter to ensure the safety of digoxin in this case?

- A. Liver function test (ALT)
- B. Serum electrolyte (potassium)
- C. Thyroid function test (thyroxin)
- D. Complete blood count (hemoglobin)

ANSWER: B

49. A 20-year-old female, who is on combined oral contraceptive for the last seven months, came to an emergency department with a chief complaint of right upper leg swelling and tenderness. She was diagnosed with deep vein thrombosis and prescribed warfarin therapy after completing her acute phase treatment.

What is the most appropriate monitoring parameter for warfarin therapy in this case?

- A. D-dimer test
- B. Prothrombin time
- C. International normalized ratio
- D. Activated partial thromboplastin time

ANSWER: C

50. A 25-year-old woman with known bronchial asthma came to OPD with chief complaints of shortness of breath and cough of two days duration. She is a resident of a rural area and lives with her cattle in the same house; which is too narrow and lacks windows. She has been using wood for cooking food in this house. She has no other pertinent history. She was on her salbutamol puff as needed. However, she had difficulty of controlling her symptoms with salbutamol.

What is the most likely social history that is triggering her asthma in this case?

- A. Salbutamol puff
- B. Being from rural
- C. Tobacco smoking
- D. Animal dander and wood smoke

ANSWER: D

51. An 11-year-old boy is presented to OPD with chief complaints of shortness of breath and cough that exacerbates at night. He has trouble of breathing two to three times per week, starting from the last two weeks. He has a wheezing sound and oxygen saturation is 85 percent on physical examination. Complete blood count is elevated with eosinophilia. Now, he has been diagnosed with acute severe bronchial asthma.

What is the most likely investigation finding that suggests asthma diagnosis in this case?

- A. Elevated CBC
- B. Increased cough severity
- C. Increased trouble breathing
- D. Reduced oxygen saturation to 85 percent

ANSWER: D

52. A 17-year-old child is presented to OPD with chief complaints of constant sneezing, runny nose, watery and itchy eyes, as well as troubling type of nasal congestion. He was diagnosed as having allergic rhinitis for which non-pharmacologic therapy was ineffective.

What is the most appropriate initial pharmacologic intervention in this case?

- A. Oral corticosteroid
- B. Parenteral corticosteroid
- C. Intra nasal corticosteroid
- D. Intra nasal steroid with oral corticosteroid

ANSWER: C

53. A 45-year-old man with known chronic obstructive pulmonary disease on salbutamol puff as needed and tiotropium puff daily therapy came to a clinic for routine follow-up with good disease control. What is the most appropriate parameter for checking effectiveness in this case?
- A. Annual chest X-ray
 - B. Annual CBC measurement
 - C. Annual arterial blood gas measurement
 - D. Annual spirometer measure of lung function test
- ANSWER: D
54. A prescriber wants to prescribe diazepam 5mg PO QD for a patient. What is the most appropriate means the prescriber must use to order this drug?
- A. Verbal request
 - B. Normal prescription paper
 - C. Ordinary prescription paper
 - D. Special prescription paper
- ANSWER: D
55. A patient came to a community pharmacy with a prescription requesting 50mg indomethacin P.O BID for one month. The pharmacist advised the patient to take the current medication thirty minutes after a meal. After taking the medication for a week, the patient came back complaining severe gastric ulcer. Earlier, the patient has no history of gastric ulcer. What is the most likely cause of gastric ulcer in this scenario?
- A. Prescribing error
 - B. Dispensing error
 - C. Administration error
 - D. Drug-food interaction
- ANSWER: C
56. A hypertensive pregnant woman came to a pharmacy with a prescription requesting amlodipine 2.5mg PO QD and atorvastatin 10mg PO QD for a month. A pharmacist reviewed the prescription, contacted the prescriber, and recommended regimen modification. What is the most likely cause of recommendation?
- A. Under dose
 - B. Drug duplication
 - C. Contraindication
 - D. Drug-drug interaction
- ANSWER: C
57. A patient came to a community pharmacy with a prescription requesting morphine 15mg IM TID. A community pharmacist received the prescription and reviewed it. What is the most likely initial step the pharmacist should do?
- A. Ensure the validity of the prescription
 - B. Counseling patients about drug side effects
 - C. Check the availability of medication and its dose
 - D. Contact the prescriber to recommend safe medication
- ANSWER: A

58. A pharmacist received a prescription calling for ceftriaxone 50 mg/kg/day in two divided doses for 10 days from a pediatric ward nurse. The patient is a six-year-old, weighing 20 kg and diagnosed with meningitis. The medication is available in a vial containing 1g powder for reconstitution. What is the appropriate number of vials to be dispensed as per the prescription?

A. 5
B. 10
C. 15
D. 20

ANSWER: B

59. A 67-year-old male patient, who is on follow-up at the anticoagulation clinic, came to a pharmacy with a prescription containing Warfarin 5mg PO one tablet taken hs. What is the most appropriate interpretation of hs in this case?

A. It means after meals
B. It means before meals
C. It means at bedtime
D. It means in the morning

ANSWER: C

60. A pharmacist received a prescription calling for phenobarbital 60 mg PO QD and chloramphenicol 500 mg IV QID. The pharmacist contacted the prescriber and recommended treatment modification. What is the most likely reason the pharmacist contacted the prescriber in the above case?

A. Drug-food interactions
B. Drug-drug interaction
C. Wrong medication strength
D. History of poor patient compliance

ANSWER: B

61. A pharmacist received a prescription requesting 375mg amoxicillin/clavulanate tablet PO TID for 10 days for a two-year-old child. The pharmacist reviewed the prescription and identified an error that needed correction.

What is the most likely prescribing error in the prescription?

A. Dosage form
B. Contraindication
C. Duration of treatment
D. Frequency of administration

ANSWER: A

62. A pharmacist received a prescription calling for 120 tablets of 30 mg of prednisolone. The pharmacist wants to count the tablets and repack them in labeled plastic containers. What is the most appropriate dispensing aid the pharmacist should use?

A. Spatula
B. Balance
C. Pan weighing scale
D. Measuring cylinder

ANSWER: C

63. A 45-year-old hypertensive patient received enalapril 10 mg PO BID for 10 days and came to a community pharmacy complaining of hypotension and persistent dry cough. What is the most likely cause of the patient's complaint?

A. Drug interaction
B. Wrong medication
C. Medication adverse effects
D. Poor response due to underdose

ANSWER: C

64. A 45-year-old female HIV patient came to ART pharmacy to collect the newly prescribed trimethoprim-sulphamethoxazole (160/800) tablet P.O daily for opportunistic infection prophylaxis. What is the most appropriate counseling tip for the patient to prevent crystalluria?

A. Break the tablet and take in a divided dose
B. Maintain adequate fluid intake
C. Decrease the dose by 50 percent
D. Take early in the morning

ANSWER: B

65. A client having eczema has come to a pharmacy with a prescription from a dermatologist requesting extemporaneous 5 gm sulphur ointment to be applied on the affected area twice daily for two weeks. The prescriptions have been stated as; Salicylic Acid BP 2 percent, Sulphur BP 3 percent, and Petrolatum; Q.s to prepare 5 g of ointment.

What is the amount of salicylic acid required to prepare the requested ointment?

A. 50 mg
B. 100 mg
C. 150 mg
D. 200 mg

ANSWER: B

66. A mother came to a pharmacy with her seven-year-old daughter for advice regarding her daughter's case. The mother observed a small motile roundworm on the daughter's faeces since yesterday. On further assessment, the pharmacist confirmed it is intestinal ascariasis otherwise no other health-related concern.

What is the most appropriate treatment of this case based on the pharmacist's assessment?

A. Tinidazole 1g P.O stat
B. Albendazole 400 mg P.O stat
C. Amoxicillin 250 mg P.O TID for 7 days
D. Metronidazole 250 mg P.O TID for 7 days

ANSWER: B

67. A 20-year-old third year male pharmacy student came to a pharmacy requesting an OTC medication of guaifenesin syrup. The pharmacist assessed the student with a mild dry cough due to a common cold.

What is the most appropriate decision that would help to respond to the symptoms?

A. Refer to a health facility
B. Deny to dispense any medication
C. Replace with dextromethorphan syrup
D. Dispense guaifenesin syrup as per his request

ANSWER: C

68. A pharmacist working in the urology department ward pharmacy wants to arrange fast-moving medical supplies close to the dispensing counter to ease frequent dispensing.

What is the most likely medical supply that has to be arranged close to the counter in this case?

- A. Catheter and urine bag
- B. Glove and elastic bandage
- C. Syringe and colectomy bag
- D. Catheter and insulin syringe

ANSWER: A

69. A patient came to your pharmacy with a valid prescription containing Mebendazole 100 mg capsule intended to be taken orally BID for three days.

What is the daily dose of Mebendazole the patient has to take?

- A. 100 mg
- B. 200 mg
- C. 300 mg
- D. 600 mg

ANSWER: D

70. A patient was prescribed Nystatin suspension for the treatment of candidiasis. The order written on the prescription showed an oral dose of 500,000 units every six hours for five days. A suspension containing 100,000 units/mL is available at the pharmacy.

What volume of suspension has to be dispensed by the pharmacist?

- A. 50 mL
- B. 100 mL
- C. 150 mL
- D. 200 mL

ANSWER: B

71. A physician prescribes a combination of amoxicillin and clavulanate suspension for a child to take 5ml orally twice a day for seven days. The child's mother came to a pharmacy and handed over the prescription paper to a pharmacist. The pharmacist found out that the strength of the suspension was not mentioned.

What should be the most appropriate option to solve the problem in this case?

- A. Consulting his colleagues
- B. Contacting the physician
- C. Specifying the suspension strength
- D. Dispensing the available suspension

ANSWER: B

72. A client having dermatitis came to a pharmacy with a prescription from a dermatologist requesting extemporaneous 30 gm medicated cream to be applied on the affected area twice daily for two weeks as follows; Salicylic Acid BP 2 percent, Sulphur BP 3 percent and Nivea cream; Q.s to prepare 30 gm of aqueous cream.

What is the appropriate duration of time bound the use date for this cream?

- A. 1 week
- B. 4 weeks
- C. 8 weeks
- D. 16 weeks

ANSWER: B

73. A 26-year-old newly diagnosed type one diabetic patient came to a hospital pharmacy with a legal prescription from a diabetic clinic to collect insulin injections of three vials.

What is the most appropriate labeling for the storage condition of the insulin vials?

- A. Store all three vials in the refrigerator
- B. Store all three vials at room temperature
- C. Store two vials at room temperature and the already started one in a refrigerator
- D. Store two vials in the refrigerator and the already started one at room temperature

ANSWER: D

74. A hospital nurse came to a surgical ward pharmacy to collect orders for wound care. The patient has heavy exudative wound due to a car accident occurred two days ago. The pharmacist dispensed the right dressing for the patient based on the order.

What is the most appropriate type of dressing required for this patient?

- A. Film Dressings
- B. Adhesive plaster
- C. Absorbent cotton
- D. Impregnated Gauze

ANSWER: C

75. A mother took her five-year-old child to a nearby clinic with complaints of fever, watery diarrhea, and nausea for two days duration. A physician diagnosed the child with acute gastro enteritis and prescribed cotrimoxazole 240mg/5ml suspension to be taken as 5ml P.O BID for five days.

What is the most appropriate patient counselling in this case?

- A. Take it before meal
- B. Take it with fatty meal
- C. Shake it well before use
- D. Avoid eating cheese and eggs

ANSWER: C

76. A 55-year-old male patient with an angina pectoris is taking metoprolol 50mg BID and nitroglycerine 0.5mg PRN. Today, the patient came to a health center and complained about erectile dysfunction and a health officer prescribed sildenafil 25mg to be taken one hour before sexual activity. The pharmacist refuses to dispense the medication although the medication is available.

What is the most likely reason for NOT dispensing sildenafil?

- A. It causes hypertension
- B. It is contraindicated for the elderly
- C. It is contraindicated with metoprolol
- D. It is contraindicated with nitroglycerine

ANSWER: D

77. A 20-year-old year-two university student started fluoxetine 20 mg P.O daily for the diagnosis of generalized anxiety disorder. After a week, the patient came to the pharmacy and informed that he decided to discontinue the medication as there was no significant improvement although he took the medication appropriately for a week.

What is the most appropriate advice for this patient?

- A. Recommend to discontinue the medication
- B. Increase the dose by making 20 mg P.O BID
- C. Change it with amitriptyline 25 mg P.O daily
- D. Continue the medication to achieve optimal improvement

ANSWER: D

78. A 30-year-old patient came to a community pharmacy complaining of orange discoloration of urine since today without any other urinary symptoms. Up on the pharmacist's assessment, the patient started four drugs regimen of anti-tuberculosis medication this morning from the nearby health center.

What is the most appropriate advice the pharmacist could give to the patient?

- A. Refer to health facility as it may be severe urinary tract infections
- B. Recommend Ciprofloxacin 500 mg tablet P.O BID for 7 days
- C. Assure the patient that it could be due to the medications
- D. Stop taking the medication and resume after a week

ANSWER: C

79. A 32-year-old female patient had a clean major surgery three days ago for myomectomy. The patient is stable except for post-operative pain and hence, the surgeon discharged her with diclofenac 50 mg tablet P.O PRN until her next week's appointment. On the date of discharge, she experienced Epigastric pain after taking diclofenac 50 mg ingestion and came to a pharmacy for a possible solution.

What is the most likely advice of the pharmacist to solve her problem?

- A. Decrease the dose to 25 mg
- B. Change the tablet to a suppository
- C. Change to aspirin 500 mg tablet P.O PRN
- D. Change to Ibuprofen tablet 400 mg P.O PRN

ANSWER: B

80. A mother came to a pharmacy to buy a cough suppressant for her six-year-old son. She informed the pharmacist that he has high-grade fever, productive cough, and chills for the past three days.

What is the most appropriate management for her query?

- A. Refer her to visit a clinic or a hospital
- B. Re-assure the patient that it will be fine by it self
- C. Dispense Chlorpheniramine 2mg/5ml Syrup, 5 ml P.O TID for 7 days
- D. Dispense Amoxicillin 250mg/5ml suspension, 5 ml P.O TID for 7 days

ANSWER: A

81. A female client received betamethasone 0.1 percent cream and 0.1 percent ointment from a tertiary care hospital pharmacy for newly diagnosed vitiligo, which involves part of her face and neck. To start the medication, the patient was confused and forgot how to take the medications and came to a nearby pharmacy.

What should be the most likely instruction on how to apply the medications?

- A. No difference applies as you want
- B. Apply the cream first and then continue with the ointment
- C. Apply the cream on the face and ointment on the neck
- D. Apply the ointment on the face and the cream on the neck

ANSWER: C

82. A junior pharmacist received a prescription ordering amoxicillin+clavulanic acid 312.5 mg/5ml suspension, 5 ml P.O TID for 7 days. The medication was available in powder form to be reconstituted immediately before use. The pharmacist counselled the patient, including storage conditions based on the patient's information leaflet, which states, "store at a temperature of 2 to 8 degrees centigrade after reconstitution".

What is the best storage condition to counsel the patient for this product at home?

- A. On a frozen (frigid)
- B. In a refrigerator on the upper shelf
- C. In a refrigerator on any lower shelf
- D. At room temperature with no direct sunlight

ANSWER: C

83. A pharmacist received a prescription calling for silver sulfadiazine cream one percent daily application on infected burns.

What most appropriate medical equipment should be provided with the medication?

- A. Atomizer
- B. Sterile applicator
- C. Suture and needles
- D. Bottle filled with sterile water

ANSWER: B

84. A nurse working in a surgery ward came to a hospital pharmacy with a prescription requesting a reusable syringe and needle. The nurse consulted the pharmacist about a decontamination procedure before reuse.

What is the most recommended decontamination procedure in the above case?

- A. Cleaning
- B. Disinfection
- C. Sterilization
- D. Boiling at 60 degrees centigrade

ANSWER: C

85. A pharmacist working in a community pharmacy arranged all his dispensing products, including medicines and medical supplies. After arranging the products, the pharmacist ordered the items using first-in, first-out criteria based on their expiry date to prevent product expiry and related wastage.

Which medical supply most likely needs more attention while arranging items accordingly?

- A. Cotton
- B. Thermometer
- C. Elastic bandage
- D. Pregnancy test kits

ANSWER: D

86. A pharmacist working in his own pharmacy ordered pregnancy test kits from a supplier. Upon arrival of the ordered items, the pharmacist carefully checked and documented all the necessary information about the kits.

What most important information need to be included to provide quality pharmacy service?

- A. length of the kit, batch number, and expiry date
- B. packing labels, batch number, and expiry date
- C. packing labels, weight of the kit, and expiry date
- D. packing labels, batch number, and weight of the kit

ANSWER: B

87. A 48-year-old male diabetic patient came to a pharmacy requesting a glucometer to monitor his serum glucose at home. A pharmacist checked the specifications of the available glucometer and dispensed it accordingly.

What are the most important parameters of specification to be checked in this case?

- A. Life of the battery and size of screen
- B. Measurement range and size of screen
- C. Precision, accuracy and measurement range
- D. Color of the strip, accuracy, and measurement range

ANSWER: C

88. A newly graduated pharmacist was assigned in a hospital pharmacy as store manager. On his first day of work, disposable gloves were received from a supplier, and the pharmacist is expected to document the transaction in a separate sheet based on the type of items.

What is the most appropriate category to be used for the documentation of the above item?

- A. Medicine
- B. Medical device
- C. Medical supplies
- D. Medical equipment

ANSWER: C

89. A 48-year-old male engineer came to a pharmacy to check his blood pressure. He was diagnosed with hypertension for four years ago and has been taking nifedipine 10 mg P.O. BID. The engineer told the pharmacist that his physician needed a daily blood pressure record for his next appointments and he wanted to buy the instrument to measure it himself at home.

What is the most appropriate decision of the pharmacist on the client's request?

- A. Refer him to contact his physician
- B. Deny to dispense the blood pressure apparatus
- C. Dispense and demonstrate digital blood pressure apparatus
- D. Dispense and demonstrate the manual apparatus with stethoscope

ANSWER: C

90. A pharmacist working in a hospital medical device warehouse updated the checklist of the expiry dates of the items. Accordingly, the products were distributed to each ward pharmacy based on their expiry date to avoid wastage. Besides, the pharmacist reported a few quantities of expired items for his manager to dispose of.

What types of items most likely have a high risk of expiry before use in this case?

- A. Medical equipment
- B. Biological products
- C. Laboratory machines
- D. Powder for laboratory reagents

ANSWER: B

91. A drug information pharmacist plans to provide health education to the public. To determine the appropriate topic, need assessment was conducted and found that most patients' drug use was irrational. Accordingly, the pharmacist decided the ultimate topic for education to be 'rational drug use'.

What is the most appropriate yardstick that helped the pharmacist to decide the ultimate topic?

- A. The severity of the problem
- B. Prevalence of the problem
- C. Global alert on the issue
- D. His interest in the topic

ANSWER: B

92. A patient came to a pharmacy to confirm the news story on BBC that stated all patients should stop taking their "Coumadin" and switch to the "new Coumadin". She tells a pharmacist that she is not willing to take it anymore and wants the physician to find a different medication. In response to her request, the pharmacist found a copy of the news story that this patient saw to help her better understand what the news reporter had conveyed.

What most appropriate category of drug information source is this?

- A. Primary
- B. Secondary
- C. Alternative
- D. Customer information

ANSWER: C

93. A 25-year-old female patient who wants a reliable medication information source to manage her ache approached a pharmacist. She told the pharmacist as she browsed several online internet sources but suspected the dependability of those sources.

What is the most reliable online drug information source?

- A. Online source from pharmaceutical manufacturers (.com)
- B. Online source from professional associations (.org)
- C. Online promotional materials
- D. Online selling ideas

ANSWER: B

94. A pharmacist was asked to come up with evidence-based recommendations for considering a newly registered and marketed antihypertensive drug to consider in a specific hospital formulary.

What most important aspect of the medication should be addressed in this case?

- A. Compare new medication with other alternatives
- B. Use information from promoters of that medication
- C. Use prescriber experience of the medication
- D. Assess sources of the medication online

ANSWER: A

95. A drug and therapeutic committee of a hospital wants to consider a new drug in the hospital formulary. The committee requested the drug information pharmacist to evaluate the new medicine for the formulary. The pharmacist responded to the request by preparing a monograph of the medication.

What is the most likely information to be included in the summary page of the monograph?

- A. Drug interactions that are not clinically significant
- B. Complete pharmacology of medication product
- C. Important aspects of the drug product
- D. Record of all formulary considerations

ANSWER: C

96. A health care provider working in a hospital requires a series of new medication information updates. As part of his professional responsibility, the drug information pharmacist was requested to prepare pharmacotherapy updates periodically for use by health care providers. What is the most appropriate method the pharmacist should use in this case?

- A. Newsletters
- B. Research articles
- C. Drug monographs
- D. Patient information sheet

ANSWER: A

97. Drug information pharmacist developed educational material on proper handling and injection of insulin for patients. The target audiences have low literacy to comprehend the written material. What is the most appropriate means of addressing the information to patients?

- A. Posting on the hospital DIS website
- B. Distributing patient education material to patients
- C. Posting written material in the patient's waiting area
- D. Platform presentation using audiovisual materials

ANSWER: D

98. The DIS pharmacist prepared patient education and delivered a presentation on rational use of antibiotics. Then after, the pharmacist documented all the activities of the session to prevent duplication activities.

Which most appropriate element of the record prevents duplication of the same task?

- A. Time of delivery
- B. Date of the event
- C. Location of the event
- D. Specific topics covered

ANSWER: D

99. A 30-year-old male patient came to a community pharmacy and asked a pharmacist whether or not diclofenac used for headaches induces peptic ulcer disease. The pharmacist tried to probe the patient to determine the ultimate drug information needed.

What is the most pertinent background information that should be asked in this case?

- A. Who is the requestor?
- B. What responses are expected?
- C. What references should be used?
- D. Why is the requestor asking the question?

ANSWER: D

100. A 50-year-old diabetic male patient was initiated on amitriptyline for treatment of neuropathic pain. He is working as a machine operator in an organization. He came to pharmacy complaining of being sleepy at work and asked whether the new medicine he started is causing the sedation.

What is the most likely drug information category requested in this case?

- A. Drug administration
- B. Adverse drug reaction
- C. Poisoning and toxicology
- D. Indication and therapeutic use

ANSWER: B

101. A client asked a community pharmacist regarding the safety and effectiveness of the Covid-19 vaccines. The client is literate and wants to have further reading on the topic. What is the most appropriate drug information source for consumers?
- A. Centers for Disease Control and Prevention (<http://www.cdc.gov>)
 - B. Complementary and Integrative Health (<https://nccih.nih.gov>)
 - C. KidsHealth.org (<https://kidshealth.org>)
 - D. Safe Medication (<http://www.safemedication.com/>)
- ANSWER: A
102. A randomized control trial was conducted to measure the effect of using atorvastatin on the incidence of death in post-myocardial infarction patients. The investigators used a placebo for a control group and atorvastatin for an interventional group. The trial was conducted in a well-controlled, double-blind protocol. The investigators claimed atorvastatin significantly reduced the incidence of mortality. What is the most likely flaw of this study?
- A. Double-blinded studies are generally poor in quality
 - B. It is impossible to assess the internal validity of the study
 - C. RCTs are not strong for assessing cause and effect in MI patients
 - D. Denying standard therapy may be unethical for MI patients
- ANSWER: D
103. A clinical pharmacist on medical ward round was asked to provide drug information requested by a patient admitted with a diagnosis of heart failure. The clinicians requested the need to add digoxin to patient's standard therapy. After searching relevant sources of information, the clinical pharmacist has planned to respond to the request.
- What are the most important factors to be considered when formulating responses?
- A. Disease incidence, severity, duration, and prognosis
 - B. Primary, secondary, and tertiary literature resources
 - C. Identity, question category, documentation, and follow-up
 - D. Patient, disease, medication, and pertinent background
- ANSWER: D
104. A 27-year-old woman who was on sumatriptan has recurrent attacks of migraine during her menstrual period. The pharmacist requested additional medication and recommended prophylactic diclofenac after searching appropriate sources. The pharmacist wanted to know whether or not the recommendation is appropriate and its outcome.
- What is the appropriate component of documentation to be included for this purpose?
- A. Documenting ultimate question
 - B. Evaluating the materials searched
 - C. Conducting follow-up monitoring
 - D. Reporting the response to responsible body
- ANSWER: C

105. A physician working on the medical ward requested drug information regarding the use of warfarin for DVT prophylaxis. The pharmacist approached the query appropriately based on the modified systematic approach.

What is most likely responsibility of the drug information pharmacist in providing a response?

- A. Providing the physician with ready access to electronic resources
- B. Coordinating purchase of appropriate drug information resources
- C. Educating the physician about appropriate drug information resources
- D. Ensuring that drug information is correctly interpreted and applied

ANSWER: D

106. A scientist, who is about to discover a new drug, selected a gene product from an organism and underwent different tests and evaluations as an initial step of his drug discovery and development process.

What is the most likely initial phase in this scenario?

- A. Target validation
- B. Lead identification
- C. Target identification
- D. Candidate selection

ANSWER: C

107. A scientist is designing and developing a dosage form for a certain drug substance. Though the drug has good membrane permeability, the scientist is being challenged by the poor aqueous solubility of the drug substance. Thus, prior to the development of the drug to a certain dosage form, the scientist has planned to work out the solubility-enhancing strategies.

What biopharmaceutical classification system best explains this case?

- A. Class I
- B. Class II
- C. Class III
- D. Class IV

ANSWER: B

108. A pharmacist working in a pharmaceutical company prepared a tablet dosage form of a drug to be administered orally, but the drug is unstable in the gastric environment. Then he decided to coat the drug in a way that prevents drug release in the stomach but gets released in the small intestine.

What is the best enteric coating material to be applied in this case?

- A. Polyethylene glycol
- B. Hydroxypropyl cellulose
- C. Hydroxypropyl methylcellulose
- D. Hydroxypropyl methylcellulose phthalate

ANSWER: D

109. Assume you are in a manufacturing stage of empty hard gelatin capsules, and you observed that a certain length of the metal pins is introduced into the chamber containing gelatin solution and covered by the gelatin solution.

What is the most likely process used in the preparation of empty hard gelatin capsules in this case?

- A. Joining
- B. Dipping
- C. Spinning
- D. Trimming

ANSWER: B

110. A pharmaceutical manufacturing company received an API raw material purchased from foreign suppliers. Hence, the quality control experts in the company took a sample of the raw material, analyzed and compared it with the reference standard, and the aim of the test was to verify whether the supplied API raw material is really the type that the company has requested.

What quality control parameter is most likely performed in the above case?

- A. Purity
- B. Assay
- C. Identity
- D. Microbial limit

ANSWER: C

111. An administered dose of a given drug is completely dissolved in fluids in the stomach, which is assumed to be 250 mL (50 mL basal level in the stomach plus administration of the solid dose with 200 mL of water), within a pH range of 1 to 8. However, its intestinal permeability was found to be low.

What is the most likely biopharmaceutical classification system of this drug?

- A. I
- B. II
- C. III
- D. IV

ANSWER: C

112. A pharmacist working in a pharmaceutical company is engaged in the production of an oral tablet formulation that contains a drug substance that is unstable in the gastric environment. Once the pharmacist is done with the compression and has formulated a complete tablet, he has a plan to go to the next phase that makes the tablet resist its drug release in the acidic environment and get released in the alkaline one.

What is the most likely step in this case?

- A. Coating
- B. Blistering
- C. Packaging
- D. Granulation

ANSWER: A

113. A "Good Manufacturing Practice" inspector goes to inspect a pharmaceutical company. During his inspection, he observed that the toilets and cafeterias are in close proximity to the main production area. Then, the inspector took a corrective measure and forwarded his recommendation to the concerned bodies based on the "Good Manufacturing Practice" principles and procedures.

What is the most likely breached "Good Manufacturing Practice" principle in this scenario?

- A. Suitable equipment and services
- B. Approved procedures and instructions
- C. Appropriately qualified and trained personnel
- D. Adequate design and location of premises and spaces

ANSWER: D

114. An industrial pharmacist working in a pharmaceutical manufacturing company produced certain batches of solid dosage form and evaluated them as per the specifications of a given pharmacopeia. Then, the industrial pharmacist stored the product in a temporary place until the results of the tests are released from the concerned body before its final release.
What is the most likely storage place for the product in this case?
- A. Packaging area
 - B. Quarantine area
 - C. Raw material warehouse
 - D. Finished product warehouse
- ANSWER: B
115. A pharmacist, who is working in a small-scale production unit of a pharmaceutical company, is going to prepare a medicated semisolid dosage form that contains moisture-sensitive API. In this case, percutaneous absorption of the API is not needed, but only its topical effect.
What is the most likely semisolid dosage form in this case?
- A. Ointment
 - B. Cream
 - C. Paste
 - D. Gel
- ANSWER: A
116. A pharmacist has formulated an API into a semisolid dosage form for its local effect. In addition to fulfilling the basic requirements, the dosage form was prepared following the strict USP specifications of sterility, metal particles, and packaging.
What is the most likely route of administration in this case?
- A. Rectal
 - B. Vaginal
 - C. Ophthalmic
 - D. Transdermal
- ANSWER: C
117. A compounding pharmacist prepared a hydrophilic petrolatum base by melting stearyl alcohol and white wax in a steam bath. Then, he added cholesterol and white petrolatum and stirred. Finally, the mixture was allowed to cool while stirring continued until it congealed.
What was the most likely unit operation a pharmacist used in this preparation?
- A. Fusion
 - B. Levigation
 - C. Trituration
 - D. Incorporation
- ANSWER: A
118. A production pharmacist planned to manufacture fluorouracil cream. The pharmacist received the required amount of fluorouracil, carbomer-940, PEG-400, propylene glycol, and stearic acid from the store.
What is the most likely cGMP principle that the pharmacist applied in the above scenario?
- A. Assuring the identity of each ingredient
 - B. Recording any necessary in-process controls
 - C. Checking containers are clean before filling
 - D. Investigating significant deviation from expected yield
- ANSWER: A

119. A pharmacist prepared a cream formulation using a small volume of oil and a large volume of water as part of internal and external phases, respectively. Thus, the pharmacist evaluated the cream by rubbing it on the skin, and the external phase was found to evaporate, leading to increased concentration of the water-soluble drug in the film of the internal phase.

What type of cream is most likely prepared by the pharmacist in this case?

- A. Cold cream
- B. Vanishing cream
- C. Multiple emulsions
- D. Water/Oil emulsions

ANSWER: B

120. A production pharmacist manufactured sulfisoxazole acetyl oral suspension (500 mg/5 mL) and requested a pharmaceutical analyst to check the product's pourability before it's packed into its final container. Then the pharmaceutical analyst checked the viscosity.

What quality parameter is most likely evaluated by the pharmaceutical analyst in this case?

- A. Assay
- B. Sterility
- C. Stability
- D. Flowability

ANSWER: D

121. A pharmaceutical company manufactured a drug product in the form of ointment. The manufactured ointment is light-sensitive. After checking its quality, the responsible pharmacist stored it till the batch was released to the market.

What is the most appropriate storage condition the pharmacist should apply in this case?

- A. Storing in refrigerator
- B. Storing in a light transparent container
- C. Avoiding exposure to direct sunlight
- D. Avoiding exposure to more than a temperature of 37 degrees centigrade

ANSWER: C

122. A pharmacist plans to manufacture an aqueous liquid preparation of erythromycin estolate. The drug has a disagreeable taste. Its highest administered dose strength dissolves with greater than 250 ml of water with a pH between 1 and 7.5.

What is the most likely liquid dosage form the pharmacist should consider in this case?

- A. Syrup
- B. Solution
- C. Emulsion
- D. Suspension

ANSWER: D

123. A pharmacist, working in a pharmaceutical manufacturing factory, prepared castor oil emulsion using granulated raw materials as per the manufacturing formula. After finishing a mixing process, the pharmacist wants to make the particle sized of the mixed components fine and similar.

What is the most appropriate equipment the pharmacist is going to use in this case?

- A. Milkshake mixer
- B. Large homogenizer
- C. Bench-type homogenizer
- D. Porcelain mortar and pestle

ANSWER: B

124. A pharmacist, working in a pharmaceutical manufacturing industry, is asked to select an appropriate packaging material for each dose of amoxicillin oral suspension manufactured in bulk. What is the most appropriate final container that should be recommended in this case?
- A. Vial
 - B. Ampoule
 - C. Collapsible tubes
 - D. Wide-mouthed bottles
- ANSWER: D
125. A production pharmacist manufactured sulfisoxazole acetyl oral suspension (500 mg/5 mL) and requested a pharmaceutical analyst to check the product's pourability before it is packed into its final container. Then the pharmaceutical analyst checked the viscosity. What most likely quality parameter is evaluated by the pharmaceutical analyst in this case?
- A. Assay
 - B. Sterility
 - C. Stability
 - D. Flowability
- ANSWER: D
126. Pharmacists, working in a local quality control department, collected propofol oil-in-water emulsion from a randomly chosen community pharmacy and checked for its quality. They found aggregation of oil droplets at the surface of the emulsion. What is the most likely physical instability seen in this product?
- A. Creaming
 - B. Coalescence
 - C. Phase inversion
 - D. Phase separation
- ANSWER: C
127. An industrial pharmacist is requested to manufacture a batch of 85 percent syrup. The pharmacist received the required amount of purified water and sucrose from a store as per the batch formula. He mixed the components following the standard operating procedure. What is the next most likely unit operation to get the final product ready to pack in this case?
- A. Dissolving
 - B. Dispersing
 - C. Size reduction
 - D. Size separation
- ANSWER: A
128. A pharmacist, working in a liquid production line, manufactured a normal saline solution as per the standard operating procedure. Then, the pharmacist wants to pack the solution in an intravenous bag which is clear in appearance and possesses excellent resistance to oils. What is the most appropriate plastic packaging material that should be used in this case?
- A. Polystyrene
 - B. Polyethylene
 - C. Polypropylene
 - D. Polyvinyl Chloride
- ANSWER: D

129. A production pharmacist, working in a liquid line of a pharmaceutical manufacturing company, manufactured a drug product in solution form, packed it into its final container, and closed with a closure system.

What is the most likely next procedure that has to be applied before releasing the product?

- A. Validate critical processes
- B. Hold the products in quarantine under appropriate conditions
- C. Keep intermediate and bulk products under appropriate conditions
- D. Carry out and record in-process controls and environmental controls

ANSWER: B

130. A pharmacist, working in a pharmaceutical manufacturing industry, is going to manufacture a drug in aerosol form. The drug is susceptible to hydrolysis and degrades at temperatures less than three degrees centigrade. After having the components of the formulation, he wants to fill them into the final container.

What is the most likely filling method he should apply in this case?

- A. Cold filling
- B. Pressure filling
- C. Gravimetric filling
- D. Constant level filling

ANSWER: B

131. An industrial pharmacist manufactured salmeterol xinafoate aerosol and requested the quality control person to measure the temperature at which the formulation ignites.

What is the most appropriate quality control parameter to be applied in this case?

- A. Flash point
- B. Spray pattern
- C. Vapour pressure
- D. Flame projection

ANSWER: A

132. A pharmacist, working in a pharmaceutical manufacturing industry, is going to manufacture an aerosol formulation. The pharmacist received all necessary raw materials from the store and wants to pack the preparation through the pressure filling process.

What is the most likely preparation condition the pharmacist should create in this case?

- A. Heat container
- B. Cool propellant
- C. Have a refrigerating system
- D. Evacuate air from container

ANSWER: D

133. A production pharmacist, working in a pharmaceutical manufacturing company, is manufacturing beclomethasone dipropionate aerosol using pressure-filling apparatus. He prepared the container at first and filled it with a product concentrate.

What is the most likely next step the pharmacist should apply in this case?

- A. Label the product
- B. Put actuator over valve
- C. Pass compressed gas into container
- D. Insert valve assembly and crimp into place

ANSWER: D

134. Ministry of Health annual review meeting on immunization found that measles vaccination coverage for all regions was low compared to the target set and a possible risk for measles outbreak in many parts of the country. The Ministry of Health of the country gave feedback to all regions to prepare an action plan that would enable them to improve the measles vaccination coverage.

What is the most likely initial step of planning used in this condition?

- A. Situation analysis
- B. Setting objectives and targets
- C. Identifying potential obstacles
- D. Analyzing and selecting critical problems

ANSWER: A

135. Poor health care financing remains a major challenge for the health system of Ethiopia. It leaves households vulnerable to misery from catastrophic health expenditures and slows progress towards achieving the millennium development goals by limiting access to essential health services among the poor.

What is the LEAST likely barrier to improving health care financing in this case?

- A. Strong reliance on out-of-pocket expenditure
- B. Low government spending on the health sector
- C. Efficient and equitable utilization of resources
- D. Poorly harmonized and unpredictable donor funding.

ANSWER: C

136. By the end of 2030 GC, Ethiopia planned to eradicate tuberculosis and introduce a new BCG vaccine that protects against both childhood and adult tuberculosis.

What is the most appropriate type of planning used in this case?

- A. Action planning
- B. Tactical planning
- C. Strategic Planning
- D. Operational planning

ANSWER: C

137. A hospital pharmaceuticals starting inventory value is 1,000,000 birr, the quantity received during the period is 3,000,000 birr and the total cost of wasted pharmaceuticals due to expiry, damage, and obsolescence is valued 50,000 birr. The regional drug regulatory authority recommended this hospital to calculate and monitor the pharmaceutical wastage rate within an acceptable World Health Organization range (less than 2 percent).

What is the most likely recommended wastage rate in this condition?

- A. 0.625 percent and acceptable
- B. 1.25 percent and acceptable
- C. 2.5 percent and unacceptable
- D. 12.5 percent and unacceptable

ANSWER: B

138. A drug and therapeutic committee of a primary hospital prepared a procurement list of cisplatin, doxorubicin, vancomycin and tamoxifen for purchase from a public supplier, and the pharmacist in charge of distribution working with the supplier approved one drug from the procurement list. What is the most likely drug approved for purchase by this hospital?
- A. Cisplatin
 - B. Tamoxifen
 - C. Doxorubicin
 - D. Vancomycin
- ANSWER: D
139. An immunization supply chain of a country selects vaccines like Penta, Rota, Measles, and Pfizer, and these vaccines need different storage temperature ranges and storage equipment while being transported and stored. What is the most likely vaccine that needs an ultralow freezer rather than a refrigerator in this case?
- A. Penta
 - B. Rota
 - C. Pfizer
 - D. Measles
- ANSWER: C
140. A pharmacist working in a cold room of a supplier records the temperature of cold products in storage and during transportation by using appropriate automatic temperature monitoring devices that are kept with the product during storage and transportation. What is the most appropriate device used in this condition?
- A. Fridge tag
 - B. Thermometer
 - C. Freeze indicator
 - D. Cold chain monitoring card
- ANSWER: A
141. A six-month-old infant was admitted to a pediatric ward and the pediatrician ordered the nurse to secure normal saline 0.9 percent with an intravenous cannula but the infant's vein was invisible. The nurse advised the pharmacy store head to select the most appropriate medical supply in this condition. What was the most likely medical supply selected in this case?
- A. Spinal needle
 - B. Scalp vein set
 - C. Oxygen nasal prong
 - D. Intravenous Cannula 22G
- ANSWER: B
142. A geriatric man with a kidney problem who is unable to control his urine complained that a nearby hospital is not supplying medical equipment that is used for collecting sudden urination. Upon request of his complaint, the drug and therapeutic committee of this hospital decided to select and purchase this medical supply. What is the most likely medical supply that will be selected in this condition?
- A. Condom Catheter
 - B. Folley Catheter
 - C. Colostomy bag
 - D. Urine bag
- ANSWER: A

143. The Ministry of Health of a country is preparing pharmaceutical budget requirements for newly constructed hospitals. The ministry aimed at estimating only budget requirements, not specific medicine quantities.

What is the most appropriate method of quantification to be used in this case?

- A. Morbidity
- B. Consumption
- C. Proxy-consumption
- D. Service-level projection of budget requirements

ANSWER: D

144. A hospital does not have either a reliable inventory or top ten disease records for the past years. The drug and therapeutic committee of this hospital wants to prepare an annual quantification document of pharmaceuticals for the next year.

What is the most appropriate method of quantification to be used in this case?

- A. Morbidity
- B. Consumption
- C. Proxy consumption
- D. Service-level projection of budget requirements

ANSWER: C

145. The reporting period of one hospital is from Ginbot 1 to Sene 30. Within this reporting period, the hospital has received 200 packs of dolutegravir 50mg tablets from a supplier, and 50 packs of this drug were dumped due to expiry. In Miazia 30, the hospital had an ending balance of 100 packs of dolutegravir. Currently, there are only 50 packs of dolutegravir in stock.

What would be the most likely calculated consumption, maximum quantity, and quantity of packs to be ordered, taking note of this condition?

- A. 200,400, and 350
- B. 200,800, and 750
- C. 250,1000, and 950
- D. 350,1400, and 1350

ANSWER: A

146. Higher-level suppliers deliver health program drugs to lower service delivery points every other month as per a minimum of two months of stock, and a maximum of four months of stock and a pharmacist working at the service delivery point assessed the stock status of ceftriaxone 1 gm injection with an average monthly consumption of 1000 vials and a current stock at hand of 5000 vials and forwarded a recommendation to facilities.

What is the most likely recommended month of stock in this condition?

- A. 3 and stocked according to the plan
- B. 0.5 and emergency order point
- C. 2 and minimum stock
- D. 5 and overstock

ANSWER: D

147. A pharmacist working in a pharmaceutical supply hub wants to determine and request the quantity of vaccine required for the coming supply period from the national supply service. In addition, this pharmacist has data on the annual coverage plan, the wastage rate of vaccines, birth registration, and population size.

What is the most likely method used to determine the quantity prior to the order?

- A. Size of immunization session
- B. Proxy consumption method
- C. Target population method
- D. Consumption method

ANSWER: C

148. A health facility's drug and therapeutic committee conducts (fast, slow and non-moving) FSN analysis of drugs within a health facility by computing the inventory turnover ratio of products, and this categorization guides the forecasting, procurement, and inventory management decisions of the health facility. The analysis shows the majority of the products have an inventory turnover ratio less than one.

What is the most likely product category that describes this condition?

- A. Fast-moving
- B. Slow-moving
- C. Non-moving
- D. Moderate-moving

ANSWER: C

149. The Ethiopian pharmaceutical supply service planned to procure products like anti-TB, anti-retroviral anti-malaria, vaccines, etc that are provided freely by external donors mobilized by the program.

What is the most likely financial source of these health program drugs?

- A. Loan from Bank
- B. Government budget
- C. Government internal revenue
- D. USAID, GAVI, World Bank, Global Fund

ANSWER: D

150. A community pharmacy, which is located in a big city is planning to reduce the storage area to minimize rental expenses. There were no recorded stockout days previously. The pharmacy uses a forced ordering delivery truck due to the availability of many suppliers in the city.

What is the most likely suitable lead time in weeks in this case?

- A. Zero
- B. One
- C. Two
- D. Four

ANSWER: A

151. A pharmacist who is working in a rural health center prepared a procurement plan. He included routine products in the list based on the supply risk category.

What is the most likely objective of the procurement plan in this case?

- A. Secure product availability
- B. Transaction cost reduction
- C. Reduce costs and find discounts
- D. Get optimal supplier and reduce costs

ANSWER: B

152. A tertiary care hospital is installing a very sophisticated laboratory machine with a closed system type of reagent. A pharmacist working in this hospital is asked to purchase reagents for the above-mentioned machine.

What is the most preferred procurement method in this case?

- A. Shopping
- B. Sole source
- C. Competitive bidding
- D. Requests for quotations

ANSWER: B

153. A hospital procurement team was asked by the hospital management to prepare appropriate procurement documents. While preparing the tender document, team members are interested to consider drug affordability and its availability from multiple suppliers.

What is the most likely information to be included in the tender document in this case?

- A. Specific delivery dates
- B. Quantities of desired products
- C. Quality assurance requirements
- D. Procurement by generic name

ANSWER: D

154. A technical manager of a private pharmacy is planning to procure medical supplies with an open tender. Unfortunately, the medical supplies are too slow-moving according to the previous consumption data.

What is the most appropriate technical specification that needs focus in the tender document?

- A. Shelf life
- B. Dosage form
- C. Generic name
- D. Package labeling

ANSWER: A

155. According to a health center pharmacy's LMIS record, the consumption of a certain laboratory reagent was 50, 45, 55 and 60 boxes for the months of September, October, November, and December, respectively. The pharmacy encountered 5, 7, and 3 stockout days in the months of September, October, and November.

What is the most likely average monthly consumption in this scenario?

- A. 50
- B. 55
- C. 60
- D. 65

ANSWER: C

156. A pharmacist working in an aid organization encounters several damaged capsules of a certain medication while receiving them from suppliers. The damage was due to excess humidity during transportation. For auditing and accountability issues, he wants to store those medications appropriately.

What is the most likely storage area in this case?

- A. Separated area
- B. Lockable cabinet
- C. The top of the shelves
- D. With adequate ventilation

ANSWER: A

157. A hospital pharmacist who is working as a store manager identified different batches of a certain medication that were purchased at different times. However, he identifies that those medications purchased earlier have a longer expiry date than those newly purchased medications and wants to prevent the possible expiry by having an appropriate storage procedure.

What is the most likely suitable procedure in this case?

- A. Arrange medicines in a manner suitable for FEFO
- B. Separate and dispose of damaged and expired drugs
- C. Avoid nearby insecticides and chemicals
- D. Organize medicines according to FIFO

ANSWER: A

158. The head of a hospital pharmacy wants to do a small audit for the transaction of medications and the expenditure related to it. In order to get more appropriate data, he prefers to use stock cards rather than bin cards.

What is the most likely parameter that interested him in this case?

- A. Expiry date
- B. Product Price
- C. Batch number
- D. Ending balance

ANSWER: B

159. A pharmacist working in a certain hospital was asked to prepare a disposal mechanism for waste materials collected from different wards. He separately collected used and damaged syringes for appropriate disposal.

What is the most appropriate disposal mechanism in this case?

- A. Landfill
- B. Recycling
- C. Autoclave
- D. Incinerator

ANSWER: D

160. A hospital pharmacy had a high consumption of a certain medication with a quantity of 600 boxes since the last procurement period. Considering this high consumption data, the pharmacist requested potential suppliers to issue 1000 boxes for the next season. Unfortunately, the supplier delivered only 800 boxes.

What is the most likely percentage that denotes the service level of the supplier in this case?

- A. 20
- B. 60
- C. 75
- D. 80

ANSWER: D

161. A medical equipment importer got a contract to deliver a newly sophisticated diagnostic imaging machine to a certain country. In order to satisfy its client, a logistics officer there wants to shorten the lead time as much as possible since the manufacturing location is abroad.

Which is the most likely mode of transportation that can shorten the lead time significantly?

- A. Air
- B. Sea
- C. Rail
- D. Land

ANSWER: A

162. A government of a country organized a vaccination task force for promoting vaccine campaigns for a week. During this process the pharmacy professionals are asked to prepare an appropriate transportation and storage zone for the hepatitis B, vaccine.
What is the most likely applicable zone in this case?
- A. Uncontrolled temperature-high security
 - B. Controlled temperature with high security
 - C. Uncontrolled temperature-normal security
 - D. Cold storage with 2-8 degrees centigrade normal security
- ANSWER: D
163. A hospital pharmacy procured different medications from a known international supplier. After the commodities were dispatched from the supplier, it encountered poor distribution management, which favored longer lead times. As a result, the hospital lost a significant amount of money.
What is the most likely economic damage that occurred in this case?
- A. Theft of stock
 - B. Misdirection of the product
 - C. Unnecessary products in stock
 - D. Commodities that expire before they are used
- ANSWER: D
164. A pharmacy head who is working at a hospital is concerned about the possibility of bribe and theft for some categories of medications in the hospital. As a result, he counts the group of medications on a daily basis and reconciles with the inventory records.
Which group of medications is most likely a concern for the pharmacist in this case?
- A. Narcotics
 - B. NSAIDS
 - C. Antitussives
 - D. Cough suppressants
- ANSWER: A
165. The marketing authorization holder applies variation to one of the registered medicines to the Ethiopian Food and Drug Authority (EFDA). The company re-submits all parts of the dossier that are affected by the variation, the type of variations made, and the category of the variation.
What is the most likely accepted variation application by EFDA?
- A. Changes in the route of administration
 - B. Change in active pharmaceutical ingredient
 - C. Changes in the pharmaceutical dosage form
 - D. Change in product labeling and packaging
- ANSWER: D
166. Inspectors from the Ethiopian Food and Drug Authority planned to conduct an inspection of a newly established manufacturing facility. The inspection comprises a full review of all aspects and components of "Good Manufacturing Practice" within a manufacturing facility.
What is the most likely type of inspection conducted in the above scenario?
- A. Special
 - B. Routine
 - C. Concise
 - D. Follow-up
- ANSWER: B

167. Inspectors conduct an audit of a pharmaceutical company on a limited number of Good Manufacturing Practice requirements selected as indicators of overall Good Manufacturing Practice performance and the identification of any significant changes that have been introduced since the last inspection.

What is the most likely type of inspection conducted by the inspectors?

- A. Special
- B. Routine
- C. Concise
- D. Follow-up

ANSWER: C

168. A joint venture pharmaceutical manufacturing company opened a medicine manufacturing plant in Ethiopia. The company operates its business activity in Ethiopia and should comply with the country's pharmaceutical regulation.

What is the most likely obligation of the company?

- A. Waived from being responsible for ensuring the quality, safety, and efficacy
- B. Importing raw materials through a port of entry regardless of authorization granted
- C. Ensuring the quality and safety of raw materials and the legality of its supplier
- D. Waived from checking the medicine produced in accordance with the appropriate GMP

ANSWER: C

169. Inspectors from the Ethiopian Food and Drug Authority planned to conduct an inspection of a newly established manufacturing facility. The inspection comprises a full review of all aspects and components of Good Manufacturing Practice within a manufacturing facility.

What is the most likely type of inspection conducted in the above scenario?

- A. Routine
- B. Concise
- C. Special
- D. Follow-up

ANSWER: A

170. A senior pharmacist is involved in the pharmaceutical wholesale business. As part of the regulatory compliance, the certificate of competence should be renewed within the stipulated time frame.

What is the most likely time frame for renewing the certificate in the scenario?

- A. One year
- B. Three years
- C. Four years
- D. Five years

ANSWER: A

171. A team of inspectors from the regulatory authority conducted an inspection in one of the local medicine manufacturing plants in Ethiopia. The inspectors observed that the company is planning to manufacture penicillin products and steroids in the same production room. The inspectors noticed that this type of production will entail cross-contamination and it indicates a violation of Good Manufacturing Practice.

What is the most likely Good Manufacturing Practice principle affected by this manufacturing plant?

- A. Premise
- B. Personnel
- C. Equipment
- D. Documentation

ANSWER: A

172. A pharmaceutical company in Ethiopia applied for renewal of marketing authorization by the Ethiopian Food and Drug Authority. The product is approved under stringent regulatory authority procedures.

What is the most likely document that might be waived by the EFDA?

- A. The latest SRA-approved product information (SmPC)
- B. Dated and signed cover letter for submission of the dossier
- C. Copy of valid GMP compliance report issued by the EFDA
- D. Complete the application form on e-RIS system using the EFDA website

ANSWER: C

173. A dossier assessor pharmacist at the Ethiopian Food and Drug Authority is assigned to review one of the submitted oral antibiotics for market approval. Upon reviewing the submitted dossier, the assessor recognized the absence of an administrative and product information module from the submitted dossier.

What is the most likely missed module in the above application?

- A. Module 1
- B. Module 2
- C. Module 3
- D. Module 4

ANSWER: A

174. A local pharmaceutical importer in Ethiopia plans to hire medical representatives (promoters). The company applied to the Ethiopian Food and Drug Authority for approval and certification of its medical representatives.

What is the most likely requirement that has to be fulfilled to certify its medical representatives?

- A. The medical representative should be qualified with one year of experience
- B. Medical representative should not be registered in the database of the EFDA
- C. Promotional certification shall be valid for five years and renew following its expiry
- D. The medical representative has to avail a signed letter from the previous work office

ANSWER: D

175. A regulatory authority understands that there is emerging evidence that drugs used in the treatment of HIV can cause hepatitis. The regulatory authority wants to know more about this ADR.

What is the most likely pharmacovigilance method the regulatory authority would choose?

- A. Intensified reporting
- B. Spontaneous reporting
- C. Cohort event monitoring
- D. Targeted spontaneous reporting

ANSWER: D

176. A pharmacist involved in sales and promotion wants to know permitted and prohibited activities in pharmaceutical advertising and promotion. To get the information, he refers to one of the regulatory documents approved by House of People's Representatives.

What is the most likely regulatory document that has to be referred to by the pharmacist?

- A. Proclamation
- B. Constitution
- C. Regulation
- D. Directive

ANSWER: A

177. A local pharmaceutical importer in Ethiopia wants to import a product called Pethidine. The importer is granted registration and market authorization certificate by the Ethiopian Food and Drug Authority.

What is the most likely requirement that has to be fulfilled by the company?

- A. A special permit for the import of Pethidine is valid for 120 days
- B. Importing the pethidine through the post office packed with other medicines
- C. Reporting within ten days of purchase order expiration, if Pethidine is not imported
- D. A special import permit from EFDA shall be a prerequisite to import the Pethidine

ANSWER: D

178. A pharmacist who works in hospital pharmacy dispensed wrong medication to the patient. The patient encountered gastro intestinal bleeding and liver damage. The patient's family charged the employer of the pharmacist (hospital) for the damage occurred to the patient. The court passed a decision stating that the hospital is liable for the negligent acts of the pharmacist.

What is the most likely category of liability breached by the pharmacist?

- A. Vicarious liability
- B. Criminal liability
- C. Product liability
- D. Civil liability

ANSWER: A

179. A junior pharmacist began his job as a clinical pharmacist at the emergency department of a hospital at 2023. During his stay, he observed that diabetic patients are admitted to the emergency room due to uncontrolled glucose levels. Thus, he decided to do research on this case.

To what most likely stage of a research process does this scenario belong?

- A. Stating objectives
- B. Topic identification
- C. Identifying the method
- D. Statement of the problem

ANSWER: D

180. A junior researcher is preparing a manuscript for publication in a reputable journal, and he sent it to a university professor for critical appraisal. After a week, the professor responded with a comment that states the research is systematic, has an indistinct study design, is logical, empirical, and replicable.

What part of the research component is most likely criticized in this case?

- A. Result
- B. Objective
- C. Conclusion
- D. Methodology

ANSWER: D

181. A mother is presented to a hospital with her one-year-old child complaining of diarrhea. The pediatrician asked her for evidence, but she was not able to remember all the details of her child's last diarrhea episode and the treatment she gave two or three months following the episode.

What is the most likely type of bias committed in this case?

- A. Recall bias
- B. Observer bias
- C. Information bias
- D. Interviewee bias

ANSWER: A

182. A hospital has 10 permanently employed pharmacists with the ages of 19, 20, 20, 21, 22, 24, 27, 27, 27, and 34 (years old). The director of the hospital wanted a report from the department of pharmacy regarding the demographic characteristics of the pharmacists. What is the most appropriate measure of central tendency in this case?
- A. Mean
 - B. Mode
 - C. Median
 - D. Percentiles
- ANSWER: C
183. A newly graduated pharmacy student presented his thesis findings using graphs. He has set the frequency distribution of the continuous variable in non-overlapping intervals and counted the number of observations that fall into each class. What is the most likely type of graph used in this case?
- A. Polygon
 - B. Box plot
 - C. Bar graph
 - D. Histogram
- ANSWER: D
184. A hospital pharmacist is doing research on rational uses of medications among patients with diabetes. Using a single population proportion formula, he enrolled a total of 300 diabetic patients for interview. Then, he selected every third patient who had been presented to the endocrine appointment room for three consecutive months and working days. What is the most likely type of sampling technique used in this case?
- A. Simple Random Sampling
 - B. Multi-Stage Sampling
 - C. Systematic Sampling
 - D. Stratified sampling
- ANSWER: C
185. A pharmacist is involved in a clinical trial for a new drug. The trial results are positive, but the pharmaceutical company sponsoring the trial pressured the pharmacist to exaggerate the benefits incurred in the research publication. What is the most ethically appropriate action that has to be taken by the pharmacist?
- A. Comply with the pharmaceutical company's request
 - B. Withdraw from the clinical trial to avoid the ethical dilemma
 - C. Refuse to exaggerate the benefits and report the true results
 - D. Consult with colleagues to gather their opinions before making a decision
- ANSWER: C

186. The pharmacist is working in an intensive care unit with a capacity of five beds, four of which are occupied. The pharmacist received a call from an emergency unit where they have two patients needing intensive care unit admission. The mother of one of those patients, whom the pharmacist knows, pleads with him to save her young son. The other patient was found to be a foreigner with no relatives in the country. The pharmacist is in a dilemma of deciding who should get medical treatment in light of scarce medical resources.

What is the most likely ethical principle that the pharmacist should uphold in the above case?

- A. Fidelity
- B. Justice
- C. Autonomy
- D. Confidentiality

ANSWER: B

187. A 25-year-old woman patient who is on a treatment for TB with 2RHZE/4RH requested to use combined oral contraceptive pills for family planning. The pharmacist in the health care team was cognizant of the effect of rifampicin on oral contraceptive pills and counseled the woman to use other contraceptive methods.

What is the most likely ethical principle practiced by the pharmacist?

- A. Justice
- B. Veracity
- C. Beneficence
- D. Non-maleficence

ANSWER: D

188. A pharmacist completed his community health rotation at a local treatment center. During his clinical day, he learned that a member of the community is being cared for by the local treatment center team for ingestion of poison. Later that evening, the pharmacist called upon people to pray for the patient's health condition and treatments on social media.

What ethical principle is most likely violated by the pharmacist?

- A. Fidelity
- B. Veracity
- C. Autonomy
- D. Confidentiality

ANSWER: D

189. A clinical pharmacist who works at the inpatient ward of a hospital entered into conflict with a senior physician on prescription, dosage, and drug-to-drug interaction on the prescription. The head pharmacist in charge tried to calm the situation.

What is the most likely conflict resolution technique used by the head pharmacist?

- A. Active listening
- B. Blaming others comment
- C. Use of threatening language
- D. Person-centered comments and criticism

ANSWER: A

190. A 70-year-old man with bone cancer visited a community pharmacy with a prescription containing morphine 30mg. The pharmacist dispensed the medicine and counseled the patient about safe drug usage with narcotics and filed the prescription.
What is the most likely duration of archiving this prescription?
- A. Two years
 - B. Three years
 - C. Four years
 - D. Five years
- ANSWER: D
191. A team of pharmacists' experts visited a health center for a routine supervision. The team requested the prescriptions filled for the past three months and started to review them against the recorded document. The experts identified that a drug was dispensed with no invoices.
What is the most likely ethical standard that is breached in this case?
- A. Ensure that medicines are those approved for free
 - B. Do not dispense prescription-only drugs without prescriptions
 - C. Be ethically free to make generic substitutions of prescription drugs
 - D. Personally fill all prescriptions or ensure appropriate filling of orders
- ANSWER: A
192. A pharmacist observes the presence of uncoordinated teamwork and conflicts among male and female pharmacists. The pharmacist realized that this issue jeopardized patient care and resulted in the cancelation of the drug education program, which used to be held every morning at the OPD.
What is the most likely barrier for effective communication in the above case?
- A. Simplicity of care
 - B. Personality differences
 - C. Generational similarity
 - D. Collaborative behavior
- ANSWER: B
193. A pharmacist working at a drug shop was dispensing a drug for the patient. In most of his interactions with patients, the pharmacist communicates with patients by name in a warm greeting, offering a smile, being attentive, and maintaining friendly eye contact.
What is the most likely communication approach used by the pharmacist?
- A. Openness
 - B. Active listening
 - C. Speaking plainly
 - D. Shared decision-making
- ANSWER: A
194. A pharmacist was dispensing drugs for a newly diagnosed hypertensive patient with counseling. While counseling, the patient asked about any food or fluid restrictions. Then, the pharmacist answered with easily understandable words and by maintaining eye contact with the patient and finally linked the patient to the OPD physician for further information about the medication.
What is the most likely relational communication the pharmacist lacked?
- A. Affect
 - B. Behavior
 - C. Engagement
 - D. Attending to need
- ANSWER: D

195. A 50-year-old female patient came to a community pharmacy with a written prescription for a drug. A pharmacist provided her the medications after following the steps of dispensing. He also counseled her about the drugs, as she should uninterruptedly take the doses even after she got healthy and not to reduce the dose in order to get an effective outcome.

What is the most likely pharmacological group at risk of resistance in this case?

- A. Anti-hypertensive
- B. Anti-psychotic
- C. Contraceptive
- D. Antibiotic

ANSWER: D

196. A 22-year-old male university student came to a community pharmacy to get a laxative for his complaint of intermittent constipation. He said that he had been consuming an unidentified substance for the last two years as a stimulant for his study. The pharmacist asked him for further clarification, and he complained of cardiac and mental health problems upon using this substance.

What is the most likely unidentified substance abused in this case?

- A. Chewing Khat
- B. Smoking cigarettes
- C. Taking narcotic products
- D. Smoking Tobacco products

ANSWER: A

197. A community pharmacist provides a voluntary health education service once a week at his nearby youth center. This week, he educated adolescents about smoking cessation intervention. In his education, he clarified nicotine replacement therapy and its dosage forms.

What is the most likely clarification given in the use of nicotine replacement therapy?

- A. He recommended using it along with smoking
- B. He stated it is available in the form of Injection
- C. He recommended it for withdrawal symptoms
- D. He explained it is a solitary accessible pharmacological treatment

ANSWER: C

198. A 30-year-old married woman was presented to a health center with a complaint of intermittent fever. She had visited a family planning clinic for Intrauterine device insertion last week. On further investigation; she has lower back pain, vomiting, fatigue and little vaginal discharges with laboratory values of elevated WBC and others are in the normal range.

What is the most likely health risk in this case?

- A. Latrogenic Infections
- B. Endogenous Infections
- C. Urinary tract infections
- D. Sexually transmitted infections

ANSWER: A

199. A 35-year-old mother, whose child is suffering from recurrent diarrhea, has come to a health center. The health educator on duty wants to provide health education about the importance of food-based oral rehydration therapies and hygiene.

What is the most likely target of health education to be applied in this case?

- A. Group
- B. Individual
- C. Population
- D. Community

ANSWER: B

200. Three men who were formerly friends met in one restaurant and discussed about irrational drug use. One of them frequently promoted the idea that antibiotics helped him to recover from all respiratory tract infections faster. The other one also stated that he did not finish his prescribed antibiotics and kept them for later use. The third one noted that he used antibiotics only for self-limiting infections.

What is the most likely risk encouraged by the attitude and practices of these three men in this case?

- A. Drug interactions of antimicrobial agents
- B. Non adherence to antimicrobial agents
- C. Resistance of antimicrobial agents
- D. Uses of antimicrobial agents

ANSWER: C

